Active sensing explains cortical variability and improves visual information

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What looks like noise may reflect active sensing

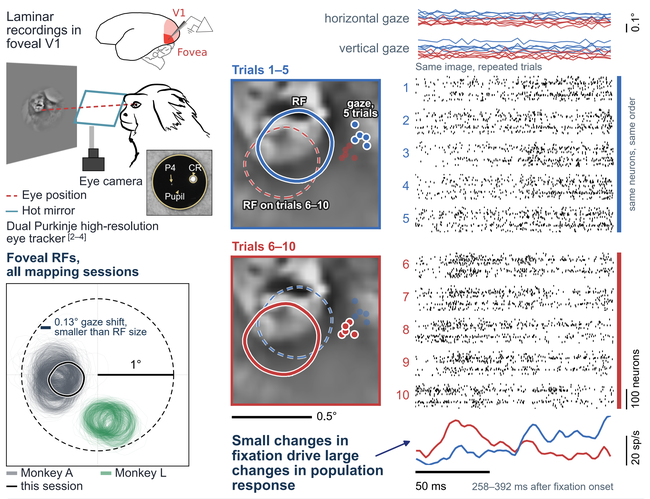
Neurons respond differently across repeated presentations of the same stimulus. These trial-to-trial differences are often described as "noise," but they can include structured responses arising from the animal's active sampling of the scene.

Even during fixation, small eye movements continually reshape visual input.

How much cortical variability comes from the eye?

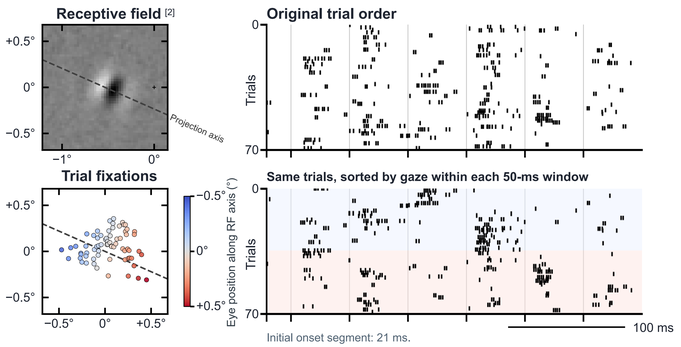
Do fixational eye movements help encode spatial detail?

1 Tiny gaze shifts reshape population responses

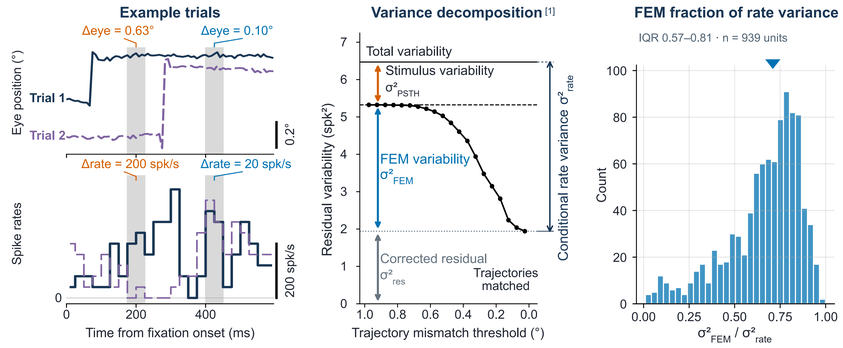


2	1,022	19	20 Hz
awake marmosets	foveal V1 units	recording sessions	repeated natural images

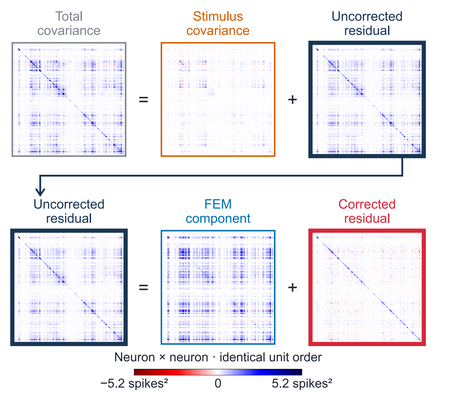
2 One neuron's "noise" tracks the eye

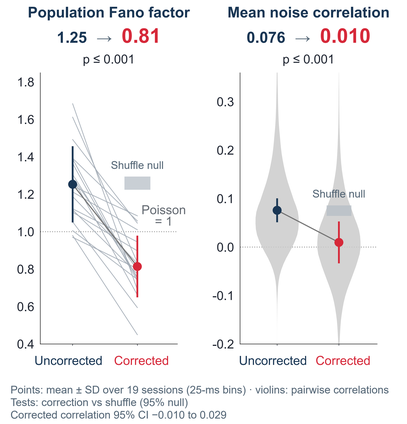


3 Eye movements drive most single-cell rate variance (71%)



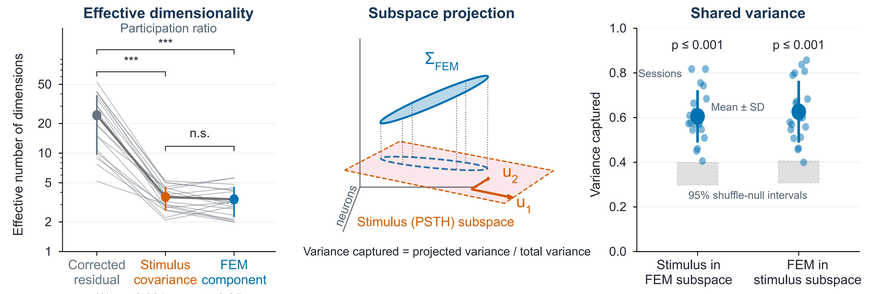
4 Correcting for gaze removes shared noise





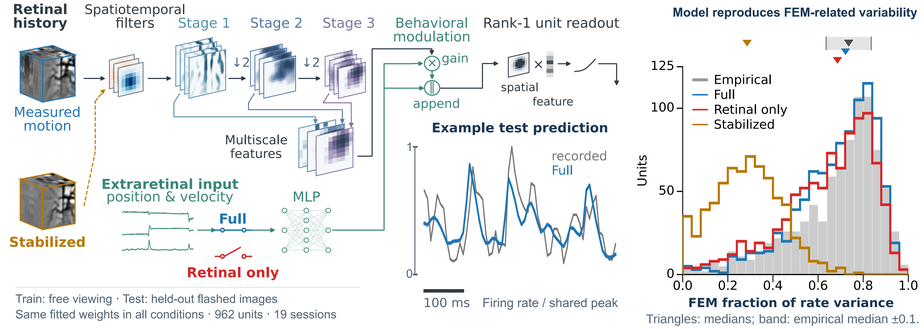
Points: mean ± SD over 19 sessions (25-ms bins) · violins: pairwise correlations
Tests: correction vs shuffle (95% null)
Corrected correlation 95% CI −0.010 to 0.029

5 Eye-driven variability follows stimulus dimensions



*** p < 0.001; n.s. p ≥ 0.05
25-ms counts; alignment uses four leading dimensions per component.

6 Gaze-dependent retinal input predicts single-trial responses



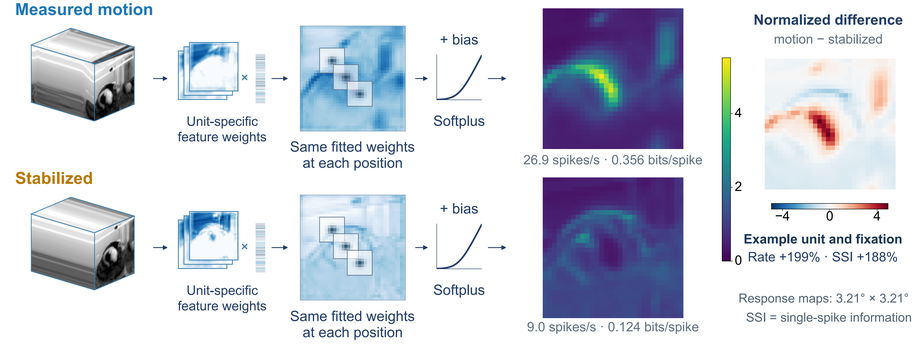
Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions · 962 units · 19 sessions

Single-trial prediction performance
Median fraction of conditional rate variance explained

Stimulus-locked average (PSTH)	Full	Retinal only	Stabilized
0.139	0.235	0.226	0.042

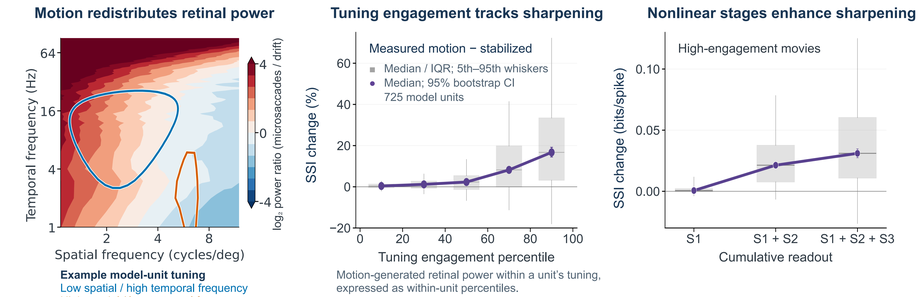
Triangles: medians; band: empirical median ±0.1.

7 In the model, eye motion sharpens spatial coding



8 In the model, sharpening increases with tuning engagement

Eye motion converts spatial structure into temporal signals that engage neuronal tuning [5,6]



Active sensing continues throughout fixation.

References
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2026 · [5] Rucci & Victor 2015 · [6] Kuang 2012

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